

Response to the Academic Editor - IoT-4522779

Manuscript IoT-4522779, MDPI *IoT*

Title: Modeling Normal Is All You Need: Joint Latent Clustering for Anomaly Detection in Multimodal Industrial-IoT Cyber-Physical Systems

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Dear Academic Editor,

Thank you for handling our manuscript. This revision addresses every request, and it does so with substantial new experimental work rather than rewriting alone: the new experiments and datasets are listed first below, followed by a short answer to each of your requests. Full point-by-point detail is in the two accompanying Reviewer letters.

New experiments and datasets added in this revision

The following are new empirical results, not present in the submitted version:

- **Computational-cost benchmark** (new Appendix B, Table B1): LatAD and the deep baselines trained and timed on one Modal A10G GPU plus a CPU edge proxy (parameters, training time, per-window latency, GPU and host memory).
- **Time-aware significance** (Section 5.4, Figure 3): every claim recomputed with an episode-block bootstrap.
- **Source-of-gain ablation** (new Table A4): cross-channel density versus a per-channel product, plus a raw-sequential-input variant and a richer temporal/spectral representation (Table A2).
- **Rare-regime imbalance sweep** (new Appendix E, Tables E1–E2) on the **newly added 3W oil-well dataset**, which has genuinely imbalanced regime occupancy.
- **Between-regime overlap (A8) screen** (new Appendix C, Table C1) across eight CPS sources, adding **Cranfield, SKAB, MetroPT, SMD, and the Paderborn bearing testbed** to WADI, HAI, and SWaT.
- **Multiscale (A9) and path-dependence (A10) tests** (Section 8): a longer-window experiment, and three conditional scorers guarded by an order-shuffle control and a synthetic positive control.
- **GDN baseline** run on all three streams under the identical protocol (Table 2, Figure 3), replacing the earlier inaccurate statement that GDN did not complete.
- **Subsystem-localization evaluation** (new Table 5), and a synthetic mechanism illustration (new Figure A1).

Corrections

We re-audited every reported number against its source artifact. The WADI difficult subset was re-derived from 43 to 30 windows (a difficulty-stratification artifact), the rescaled WADI analyzer channel was dropped, and the canonical December-2015 SWaT release was

adopted (Section 5.1); all tables were rebuilt. The main conclusions hold. The headline difficult-subset AUROC is HAI 0.845 and WADI 0.771; on the drift-dominated SWaT record the drift-aware typing of Section 7 recovers the difficult subset from 0.524 to 0.723.

Your direct requests

Please validate the assumptions that are currently unused (A3, A9, A10), provide direct experimental support for the rare-regime-safe nearest-component likelihood, report training and inference cost for edge deployment, use a time-aware statistical test, and add ablations isolating the source of the gain. Reviewer 2 additionally asks for a clearer abstract and introduction, a restructured related-work section, system-model and LatAD workflow figures, resolution of the notation conflict for z , and consistent typesetting with numbered equations. Please submit a point-by-point response with the revised manuscript.

1. Validate the assumptions that are currently unused (A3, A9, A10)

New experiments. MIIM is now ten assumptions: seven realized and validated here (A1 to A7) and three specified for future work (A8 to A10). The unused "A3" is renumbered A8 and remeasured with a new observation-space screen (Appendix C); A9 (multiscale) is shown present and partly exploitable by a longer-window test; A10 (path dependence) shows no evidence under a guarded test (three conditional scorers, order-shuffle and synthetic positive controls). See Table 1 and Section 8; full analysis in the Reviewer 1 letter (comment 1).

2. Direct experimental support for the rare-regime-safe nearest-component likelihood

New experiment and dataset. The head is isolated in Table A5, and its rare-regime behavior is measured directly on the newly added 3W oil-well dataset (Appendix E, Tables E1–E2): at a matched false-alarm rate it removes the mixture's rare-regime false alarms and stays at or below three imbalance-aware alternatives at every occupancy threshold. Full detail in the Reviewer 1 letter (comment 2).

3. Report training and inference cost for edge deployment

New experiment. Appendix B and Table B1 report parameters, training time, per-window latency, and peak GPU and host memory for LatAD and the deep baselines, on one Modal A10G GPU plus a CPU edge proxy. LatAD-global trains in 3.2 / 16 / 3.5 s on WADI / HAI / SWaT (roughly 47 to 72 times faster than TranAD) and runs within gateway-class latency and about 19 to 29 MB of GPU memory. Full detail in the Reviewer 1 letter (comment 4).

4. Use a time-aware statistical test

New analysis. Every significance claim now uses an episode-block bootstrap that resamples whole attack episodes with contiguous normal blocks (Section 5.4). On the difficult subset: HAI is significant (+0.088, $P < 0.0005$); SWaT becomes significant after the drift-aware typing of Section 7 (+0.200, $P = 0.014$); WADI is ahead of every learned detector (each $P < 0.05$) and ties only the linear baseline (8 attack episodes). Full detail, with confidence intervals, in the Reviewer 1 letter (comment 5).

5. Add ablations isolating the source of the gain

New experiment. Table A4 replaces the cross-channel latent density with a per-channel product: the cross-channel term adds +0.200 on HAI, +0.017 on WADI, and -0.050 on SWaT. The source is therefore dataset-specific (HAI cross-channel density, WADI regime-community factorization, SWaT temporal and drift structure), not the window representation. Full detail in the Reviewer 1 letter (comment 6).

Requests forwarded from Reviewer 2

These are writing, figure, and notation items; each is answered in full in the Reviewer 2 letter (comment numbers cited).

- **Abstract** (R2 1–2): abbreviations expanded at first use, and the abstract now states LatAD's mechanism before the headline result (best All-subset AUROC on HAI 0.948; difficult-subset HAI 0.845).
- **Introduction** (R2 3–5): opens with CPS/IIoT background, no contributions or method sentence in the first paragraph, the "task" named, and the two obstacles tied to contributions C1 to C3.
- **Related Work** (R2 6): definition moved to the introduction, LatAD self-descriptions removed, Section 2.3 rewritten as a neutral survey, and the old Section 2.5 replaced by a benchmark-scale section.
- **Figures** (R2 7, 9): new Figure 1 (IIoT system-model context) and new Figure 2 (LatAD workflow, each block tagged by the assumption it serves); the Section 4 headings now name each pipeline stage.
- **Notation** (R2 11): the latent keeps z , the standardization operator becomes $\text{std}(\cdot)$, and the trivial-rule per-channel value becomes u .
- **Typesetting** (R2 12): all nine display equations are numbered (1) to (9), and paragraph indentation is unified at the DOCX build stage.

Point-by-point response and revised manuscript

Please submit a point-by-point response with the revised manuscript.

This letter and the accompanying Reviewer 1 and Reviewer 2 letters are the point-by-point response. The revised manuscript is change-marked: new and changed text is highlighted, deleted passages are removed, and each response cites the section, table, or figure it changed.

Every request is addressed within the manuscript, with completed experiments and corrected numbers; the three assumptions A8 to A10 are the only items explicitly scoped to future work. Thank you again for your time.

Sincerely,
Alexander Apartsin and Yehudit Aperstein